

DOOMSDAY ALGORITHM — UI DESIGN SPECIFICATION

Neobrutalism × Marvel Comic · The Earth-4471 Archive

Data Science Club · VIT-AP · V-TAPP 2026 Target: <https://doomsdayalgorithm-5f1e8.web.app>SMARTPANTS PRESERVED 477SMARTPANTS PRESERVED 478

0. HOW TO USE THIS DOCUMENT

This is an **implementation spec, not a mood board**. Every section below is meant to be executed literally. Read sections 1, 2, and 7 before writing a single line of code — section 7 (The DOM Contract) is the one that will break the entire event if ignored.

Work in this order: §3 tokens → §4 textures → §6 components → §9 page-by-page. Do not start a page until the component library it uses exists.

Anything described as **MUST** is load-bearing. Anything described as *may* is taste.

1. STACK REALITY — READ THIS FIRST

The inspiration component you were given is **React + Tailwind + shadcn + TypeScript + framer-motion**. This codebase is **none of those things**.

	<u>Inspiration assumes</u>	<u>This project actually is</u>
<u>Framework</u>	<u>React 18, JSX</u>	<u>Vanilla DOM, <code>document.createElement</code></u>
<u>Styling</u>	<u>Tailwind 4 utility classes</u>	<u>One hand-written <code>shared/theme.css</code></u>
<u>Components</u>	<u>shadcn/ui in <code>/components/ui</code></u>	<u>No component system at all</u>
<u>Types</u>	<u>TypeScript</u>	<u>Plain ES2017 JS</u>
<u>Build</u>	<u>Vite/Next bundler</u>	<u>No build step. Files are served exactly as written</u>
<u>Deps</u>	<u>npm packages</u>	<u>Two <code><script></code> CDN tags (Firebase compat SDK)</u>
<u>Deploy</u>	<u>Node build → static out</u>	<u><code>firebase deploy --only hosting</code> on the raw folder</u>

THE RULING: DO NOT MIGRATE TO REACT.

Reasons, in order of weight:

1. **The event runs September 11, 2026.** A framework migration the day before is how a working event becomes a broken one.
2. **The zero-cost architecture depends on the current shape.** Grading runs from a laptop script against Firestore. Adding a bundler adds a build artifact that must be rebuilt and redeployed correctly under time pressure, with no CI to catch a mistake.
3. **Neobrutalism is a CSS aesthetic, not a component aesthetic.** Thick borders, hard offset shadows, flat color, zero radius — every single visual property in the reference is reproducible in plain CSS. Tailwind is a *delivery mechanism* for those properties, not a prerequisite.
4. **The interesting parts of the reference are CSS anyway.** The window chrome, the offset shadow press effect, the flat palette — none of it needs React. The one genuinely JS-driven piece (the rotating headline word) is ~15 lines of vanilla JS.

What we take from the inspiration: the windowed-panel motif, the hard-offset shadow language, the flat saturated palette, the confident oversized type, and the press-into-the-shadow interaction.

What we discard: the entire toolchain, the video background (2.3 MB over campus wifi, for a page teams look at for eight seconds), and the muted glassmorphic overlay (that is the *opposite* of brutalism — brutalism does not blur).

If someone later insists on React, that is a separate project after the event. Note it and move on.

2. DESIGN THESIS

Three ideas, layered. Every visual decision must serve at least one.

2.1 NEOBRUTALISM — *THE STRUCTURE*

Raw, honest, undecorated. Elements look like physical objects sitting on a surface, casting hard shadows. Nothing is soft, nothing fades, nothing blurs. Borders are visible and thick because the boundary between things is information. Interaction is physical: a button pressed moves *into* its shadow.

Non-negotiable rules:

- `border-radius: 0` everywhere. No exceptions. Not on buttons, not on inputs, not on cards.
- Every shadow is `Npx Npx 0` — **zero blur, zero spread**, always pure black.
- No gradients on surfaces. (One exception: the archive banner stripe, §6.2.)
- No `backdrop-filter`, no `opacity` on containers, no translucent panels.
- Every interactive element has a visible black border of at least 3px.
- Color is flat and fully saturated or it is black/paper. No tints, no 60%-opacity greys — if text needs to be quieter, use a *different flat color*, not transparency.

2.2 MARVEL COMIC — *THE SKIN*

Silver-age comic print: newsprint paper, ben-day halftone dots, panel gutters, caption boxes with hard borders, sound-effect lettering, rubber-stamp overprints.

- Panels are laid out like comic panels: rectangular, gutter-separated, irregular sizes.
- Headers behave like **caption boxes** (the yellow box in the corner of a comic panel).

- Emphasis uses **sound-effect type** (`Bangers`) sparingly — maximum one per page.
- Rubber stamps (`CLASSIFIED` , `SEALED` , `TAKEN`) are rotated 4–8° and overprinted at the element's edge.

2.3 EARTH-4471 — *THE TWIST*

This is not the real Marvel universe, and the site should feel **subtly wrong** — a corrupted archive from a parallel timeline. This is the concept that keeps the design from being generic comic-book pastiche.

Expressed through:

- **Magenta glitch accent** (`--glitch`) used *only* for wrongness: the banner, redacted values, the “not Earth-616” warning, sealed Phase 5 data.
 - **Redaction bars** — solid black rectangles over “unknown” values, especially Phase 5.
 - **Archive-file framing** — window title bars read like file headers: `EARTH-4471 // FILE: APPEARANCES.CSV` .
 - **Registration misprint** — a 1–2px magenta offset ghost behind display headlines, like badly aligned comic printing.
-

3. DESIGN TOKENS

Replace the whole of `portal/public/shared/theme.css` with this token block at the top. **Every value below is final** — do not invent new colors or spacing values while implementing.

```

/* =====
DOOMSDAY ALGORITHM – DESIGN TOKENS
Neobrutalism x Marvel x Earth-4471
===== */

:root {
  /* ---- INK & PAPER (the two structural colors) ---- */
  --ink:          #0A0A0A; /* every border, every shadow, most text */
  --paper:        #F2EDE0; /* aged newsprint – the default page ground */
  --paper-bright: #FFFBF0; /* raised surfaces: window bodies, inputs */
  --paper-dark:   #DCD5C4; /* recessed: table stripes, disabled fills */

  /* ---- MARVEL PRIMARIES (flat, saturated, never tinted) ---- */
  --marvel-red:    #ED1D24; /* primary action, danger, the brand */
  --marvel-red-deep: #B31217; /* pressed/hover state of red only */
  --marvel-yellow: #FFD400; /* caption boxes, highlights, warnings */
  --marvel-blue:   #0476D9; /* data, links, informational */
  --marvel-green:  #00A550; /* success, submitted, verified */

  /* ---- EARTH-4471 CORRUPTION ---- */
  --glitch:        #FF2E88; /* wrongness ONLY – never decorative */
  --redact:        #0A0A0A; /* redaction bars (same as ink, named for intent) */
}

/* ---- SEMANTIC ALIASES (use these in components) ---- */
--bg:              var(--paper);
--surface:         var(--paper-bright);
--text:            var(--ink);
--text-quiet:      #55504A; /* a real color, NOT ink at 60% opacity */
--accent:          var(--marvel-red);
--accent-alt:      var(--marvel-yellow);
--info:            var(--marvel-blue);
--success:         var(--marvel-green);

/* ---- BORDERS (only three widths exist) ---- */
--bw-thin:         2px;
--bw:              3px;
--bw-thick:        5px;
--border:          var(--bw) solid var(--ink);
--border-thin:     var(--bw-thin) solid var(--ink);
--border-thick:    var(--bw-thick) solid var(--ink);

/* ---- HARD SHADOWS (zero blur, always) ---- */
--sh-xs:          2px 2px 0 var(--ink);
--sh-sm:          4px 4px 0 var(--ink);
--sh:             6px 6px 0 var(--ink);

```

```

--sh-lg:          10px 10px 0 var(--ink);
--sh-xl:          14px 14px 0 var(--ink);
/* colored shadows for stacked/emphasis elements */
--sh-red:         6px 6px 0 var(--marvel-red);
--sh-yellow:      6px 6px 0 var(--marvel-yellow);

/* ---- SPACING (8px base, no arbitrary values) ---- */
--s-1: 4px;  --s-2: 8px;  --s-3: 12px;  --s-4: 16px;
--s-5: 24px;  --s-6: 32px;  --s-7: 48px;  --s-8: 64px;  --s-9: 96px;

/* ---- TYPE SCALE ---- */
--font-display: 'Archivo Black', 'Impact', sans-serif;
--font-body:    'Space Grotesk', 'Helvetica Neue', Arial, sans-serif;
--font-mono:    'JetBrains Mono', 'Consolas', monospace;
--font-fx:      'Bangers', 'Impact', cursive;

--t-xs: 12px;  --t-sm: 14px;  --t-base: 16px;  --t-lg: 19px;
--t-xl: 24px;  --t-2xl: 32px;  --t-3xl: 44px;  --t-4xl: 64px;
--t-5xl: 88px;

--lh-tight: 1.05;  --lh-snug: 1.25;  --lh-body: 1.55;

/* ---- RADIUS: THERE IS NONE ---- */
--radius: 0;

/* ---- MOTION (fast, mechanical, no easing drama) ---- */
--dur-fast: 80ms;
--dur: 140ms;
--dur-slow: 260ms;
--ease: cubic-bezier(0.2, 0, 0, 1);
--press: translate(3px, 3px);

/* ---- LAYERS ---- */
--z-base: 1;  --z-sticky: 50;  --z-nav: 100;  --z-toast: 500;  --z-modal: 900;

/* ---- LAYOUT ---- */
--w-narrow: 720px; /* report, judge */
--w-content: 960px; /* landing, question rounds */
--w-wide: 1240px; /* draft board, leaderboard */
}

```

3.1 FONT LOADING

Add to the `<head>` of **every** page, before `theme.css` :

```
<link rel="preconnect" href="https://fonts.googleapis.com">
<link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
<link href="https://fonts.googleapis.com/css2?
family=Archivo+Black&family=Space+Grotesk:wght@400;500;700&family=JetBrains+Mono:wg
rel="stylesheet">
```

`display=swap` is mandatory — a blocked font must never blank the page mid-event.

3.2 COLOR USAGE RULES (ENFORCED)

Color	Use for	NEVER use for
<code>--marvel-red</code>	Primary buttons, active nav, round numbers, danger	Body text, large fills over 30% of viewport
<code>--marvel-yellow</code>	Caption boxes, highlight bars, “your turn” markers	Text color (fails contrast on paper)
<code>--marvel-blue</code>	Data values, links, info callouts	Buttons (red owns actions)
<code>--marvel-green</code>	Submitted ✓, success toasts, verified states	Anything not-yet-successful
<code>--glitch</code>	Banner, redactions, Phase-5 sealed markers, misprint ghost	Buttons, body copy, decoration
<code>--ink</code>	All borders, all shadows, all body text	=

Contrast floor: body text on `--paper` must be `--ink` or `--text-quiet`. Never place `--marvel-yellow` text on paper. Never place `--ink` on `--marvel-red` (use `--paper-bright`).

4. TEXTURES & EFFECTS

Four recipes. Implement all four in `theme.css` as utility classes. All are pure CSS — no image assets, no downloads.

4.1 HALFTONE (BEN-DAY DOTS)

```
.tex-halftone {
  background-image: radial-gradient(var(--ink) 1px, transparent 1px);
  background-size: 6px 6px;
  background-position: 0 0;
}
.tex-halftone--red {
  background-image: radial-gradient(var(--marvel-red) 1.5px, transparent 1.5px);
  background-size: 8px 8px;
}
/* Halftone used as a fade: dots get sparse toward one edge */
.tex-halftone--fade {
  background-image: radial-gradient(var(--ink) 1px, transparent 1px);
  background-size: 6px 6px;
  -webkit-mask-image: linear-gradient(to bottom, #000, transparent);
  mask-image: linear-gradient(to bottom, #000, transparent);
}
```

Use on: hero panel background, window title bars, leaderboard header, empty states.

Opacity of the effect is controlled by dot size, never by `opacity`.

4.2 REGISTRATION MISPRINT (THE EARTH-4471 TELL)

Display headlines get a magenta ghost offset 2px up-left, like a comic printed with the plates misaligned.

```
.misprint {
  position: relative;
  color: var(--ink);
}
.misprint::before {
  content: attr(data-text);
  position: absolute;
  left: -2px; top: -2px;
  color: var(--glitch);
  z-index: -1;
  pointer-events: none;
}
```


Usage: `<h1 class="misprint" data-text="DOOMSDAY">DOOMSDAY</h1>`. The `data-text` MUST match the visible text exactly.

Apply to: the landing `<h1>` only, and the leaderboard `<h1>`. Overuse kills it.

4.3 RUBBER STAMP

```
.stamp {
  display: inline-block;
  font-family: var(--font-display);
  font-size: var(--t-sm);
  letter-spacing: 0.08em;
  text-transform: uppercase;
  color: var(--marvel-red);
  border: var(--bw) solid var(--marvel-red);
  padding: var(--s-1) var(--s-3);
  transform: rotate(-6deg);
  background: transparent;
}
.stamp--taken { color: var(--ink); border-color: var(--ink); transform:
rotate(4deg); }
.stamp--sealed { color: var(--glitch); border-color: var(--glitch); }
.stamp--done { color: var(--marvel-green); border-color: var(--marvel-green);
transform: rotate(-3deg); }
```

Use on: drafted character cards (`TAKEN`), Phase-5 references (`SEALED`), completed rounds (`DONE`).

4.4 REDACTION BAR

```
.redact {
  background: var(--redact);
  color: var(--redact); /* text present for screen readers, invisible
visually */
  user-select: none;
  padding: 0 var(--s-2);
}
.redact::selection { background: var(--redact); color: var(--redact); }
```

Use on: any Phase 5 value shown as a placeholder, the answer-key teaser on the landing page.

5. TYPOGRAPHY SYSTEM

```

body {
  font-family: var(--font-body);
  font-size: var(--t-base);
  line-height: var(--lh-body);
  color: var(--text);
  background: var(--bg);
  margin: 0;
  -webkit-font-smoothing: antialiased;
}

h1, .h1 {
  font-family: var(--font-display);
  font-size: clamp(var(--t-3xl), 7vw, var(--t-5xl));
  line-height: var(--lh-tight);
  letter-spacing: -0.02em;
  text-transform: uppercase;
  margin: 0 0 var(--s-4);
}

h2, .h2 {
  font-family: var(--font-display);
  font-size: clamp(var(--t-xl), 4vw, var(--t-2xl));
  line-height: var(--lh-snug);
  letter-spacing: -0.01em;
  text-transform: uppercase;
  margin: 0 0 var(--s-3);
}

h3, .h3 {
  font-family: var(--font-body);
  font-weight: 700;
  font-size: var(--t-lg);
  text-transform: uppercase;
  letter-spacing: 0.04em;
  margin: 0 0 var(--s-2);
}

p { margin: 0 0 var(--s-3); max-width: 68ch; }

/* Data, IDs, question codes, percentages, scores – ALWAYS mono */
code, .mono, .num { font-family: var(--font-mono); font-weight: 700; }

/* Sound effect – max one per page */
.fx {
  font-family: var(--font-fx);
  font-size: clamp(var(--t-2xl), 6vw, var(--t-4xl));

```

```

    color: var(--marvel-red);
    letter-spacing: 0.02em;
    transform: rotate(-4deg);
    display: inline-block;
  }

  /* Caption box – the yellow narration box from comics */
  .caption {
    display: inline-block;
    background: var(--marvel-yellow);
    border: var(--border);
    box-shadow: var(--sh-sm);
    padding: var(--s-2) var(--s-3);
    font-family: var(--font-body);
    font-weight: 700;
    font-size: var(--t-sm);
    text-transform: uppercase;
    letter-spacing: 0.06em;
  }

```

Rule: every number a participant is scored on renders in `--font-mono`, bold.

Percentages, ranks, totals, question IDs, point values. This makes data visually distinct from prose across the whole site.

6. COMPONENT LIBRARY

Build every one of these in `theme.css` before touching any page.

6.1 THE WINDOW (`.WIN`) — THE SIGNATURE COMPONENT

This is the motif carried from the reference. **Every content block on the site is a window.** No bare `<div>` panels.

```
<section class="win">  
  <header class="win_bar">  
    <span class="win_dots" aria-hidden="true"><i></i><i></i><i></i></span>  
    <span class="win_title">EARTH-4471 // ROUND 01 – EXPLORE</span>  
  </header>  
  <div class="win_body">  
    <!-- content -->  
  </div>  
</section>
```

```

.win {
  background: var(--surface);
  border: var(--border);
  box-shadow: var(--sh);
  margin-bottom: var(--s-5);
}
.win bar {
  display: flex;
  align-items: center;
  gap: var(--s-3);
  background: var(--ink);
  color: var(--paper);
  border-bottom: var(--border);
  padding: var(--s-2) var(--s-3);
  font-family: var(--font-mono);
  font-size: var(--t-xs);
  font-weight: 700;
  letter-spacing: 0.08em;
  text-transform: uppercase;
}
.win dots { display: inline-flex; gap: var(--s-1); flex-shrink: 0; }
.win dots i {
  width: 11px; height: 11px;
  border: var(--bw-thin) solid var(--paper);
  background: transparent;
  display: block;
}
.win dots i:first-child { background: var(--marvel-red); }
.win title { overflow: hidden; text-overflow: ellipsis; white-space: nowrap; }
.win body { padding: var(--s-5); }

/* Variants */
.win--accent .win bar { background: var(--marvel-red); }
.win--warn .win bar { background: var(--marvel-yellow); color: var(--ink); }
.win--warn .win dots i { border-color: var(--ink); }
.win--glitch .win bar { background: var(--glitch); }
.win--flush .win body { padding: 0; } /* for tables and boards */
.win--lift { box-shadow: var(--sh-lg); } /* hero / leaderboard */

```

Title bar copy convention — always `EARTH-4471 // <CONTEXT>`, uppercase, mono:

- `EARTH-4471 // BRIEFING`
- `EARTH-4471 // ROUND 01 — EXPLORE`

- [EARTH-4471 // FILE: APPEARANCES.CSV](#)
- [EARTH-4471 // DRAFT BOARD — 43 ASSETS](#)
- [EARTH-4471 // STANDINGS \[LIVE\]](#)

6.2 ARCHIVE BANNER ([.BANNER](#))

Full-bleed strip at the very top of every page. Already rendered by [shared/banner.js](#) — restyle only, do not change the JS.

```
.banner-earth4471 {  
  background: var\(--glitch\);  
  color: var\(--ink\);  
  border-bottom: var\(--border-thick\);  
  padding: var\(--s-2\) var\(--s-4\);  
  font-family: var\(--font-mono\);  
  font-weight: 700;  
  font-size: var\(--t-xs\);  
  letter-spacing: 0.1em;  
  text-transform: uppercase;  
  text-align: center;  
}
```

6.3 NAVIGATION ([.TOP-NAV](#))

Rendered by [shared/nav.js](#) — restyle only.

```
nav.top-nav {  
  display: flex;  
  gap: 0;  
  background: var(--paper);  
  border-bottom: var(--border-thick);  
  padding: 0 var(--s-4);  
  position: sticky;  
  top: 0;  
  z-index: var(--z-nav);  
  overflow-x: auto;  
}  
nav.top-nav a {  
  font-family: var(--font-display);  
  font-size: var(--t-sm);  
  text-transform: uppercase;  
  letter-spacing: 0.06em;  
  color: var(--ink);  
  text-decoration: none;  
  padding: var(--s-3) var(--s-4);  
  border-right: var(--border);  
  white-space: nowrap;  
  transition: background var(--dur-fast) var(--ease);  
}  
nav.top-nav a:first-child { border-left: var(--border); }  
nav.top-nav a:hover { background: var(--marvel-yellow); }  
nav.top-nav a.active { background: var(--marvel-red); color: var(--paper-bright); }
```


6.4 BUTTONS

```

button, .btn {
  font-family: var(--font-display);
  font-size: var(--t-sm);
  text-transform: uppercase;
  letter-spacing: 0.06em;
  background: var(--marvel-red);
  color: var(--paper-bright);
  border: var(--border);
  box-shadow: var(--sh-sm);
  padding: var(--s-3) var(--s-5);
  cursor: pointer;
  transition: transform var(--dur-fast) var(--ease),
              box-shadow var(--dur-fast) var(--ease),
              background var(--dur-fast) var(--ease);
}
button:hover, .btn:hover { background: var(--marvel-red-deep); }
button:active, .btn:active { transform: var(--press); box-shadow: var(--sh-xs); }
}
button:focus-visible, .btn:focus-visible {
  outline: var(--bw) solid var(--marvel-blue);
  outline-offset: 3px;
}
button:disabled, .btn:disabled {
  background: var(--paper-dark);
  color: var(--text-quiet);
  box-shadow: none;
  transform: var(--press);
  cursor: not-allowed;
}

/* Variants */
.btn--secondary { background: var(--paper-bright); color: var(--ink); }
.btn--secondary:hover { background: var(--marvel-yellow); }
.btn--ghost { background: transparent; color: var(--ink); box-shadow: none; }
.btn--lg { font-size: var(--t-lg); padding: var(--s-4) var(--s-7); box-shadow:
var(--sh); }
.btn--block { display: block; width: 100%; }

/* Submitted state – set by JS changing.textContent to 'Submitted ✓' */
.btn--done {
  background: var(--marvel-green);
  color: var(--paper-bright);
  box-shadow: var(--sh-xs);
}

```

```
transform: var(--press);  
cursor: default;  
}
```

The press effect is the soul of this design. transform: translate(3px,3px) +
shadow shrinking from 4px to 2px = the button physically moves into its own shadow.
Every clickable thing on the site does this.

6.5 INPUTS, TEXTAREAS, RANGE SLIDERS

```

input[type="text"], input:not([type]), textarea {
  font-family: var(--font-mono);
  font-size: var(--t-base);
  background: var(--paper-bright);
  color: var(--ink);
  border: var(--border);
  box-shadow: inset var(--sh-xs);
  padding: var(--s-3);
  width: 100%;
}
input:focus, textarea:focus {
  outline: none;
  border-color: var(--marvel-blue);
  box-shadow: var(--sh-sm);
}
textarea { resize: vertical; min-height: 72px; }

label {
  display: block;
  font-family: var(--font-body);
  font-weight: 700;
  font-size: var(--t-sm);
  text-transform: uppercase;
  letter-spacing: 0.04em;
  margin-bottom: var(--s-2);
}

/* Number inputs used for multichoice probability splits */
input[type="number"] {
  font-family: var(--font-mono);
  font-weight: 700;
  border: var(--border);
  background: var(--paper-bright);
  padding: var(--s-2);
  width: 5rem;
  text-align: center;
}

/* Range slider – the primary Predict/Explore control. Style ALL THREE
   vendor tracks or it silently falls back to the OS default on one browser. */
input[type="range"] {
  -webkit-appearance: none;
  appearance: none;

```

```

    width: 100%;
    height: 28px;
    background: transparent;
    cursor: pointer;
}
input[type="range"]::-webkit-slider-runable-track {
    height: 16px;
    background: var(--paper-dark);
    border: var(--border);
}
input[type="range"]::-moz-range-track {
    height: 16px;
    background: var(--paper-dark);
    border: var(--border);
}
input[type="range"]::-webkit-slider-thumb {
    -webkit-appearance: none;
    width: 28px; height: 28px;
    margin-top: -9px;
    background: var(--marvel-red);
    border: var(--border);
    box-shadow: var(--sh-xs);
}
input[type="range"]::-moz-range-thumb {
    width: 28px; height: 28px;
    background: var(--marvel-red);
    border: var(--border);
    box-shadow: var(--sh-xs);
    border-radius: 0;
}

/* Checkbox – judge checklist. Custom, chunky, unmistakable. */
input[type="checkbox"] {
    appearance: none;
    -webkit-appearance: none;
    width: 24px; height: 24px;
    border: var(--border);
    background: var(--paper-bright);
    cursor: pointer;
    vertical-align: middle;
    margin-right: var(--s-2);
    flex-shrink: 0;
}
input[type="checkbox"]:checked {
    background: var(--marvel-red);

```

```
box-shadow: inset 0 0 0 4px var(--paper-bright);  
}
```

6.6 QUESTION CARD (.q)

Used in Explore and Predict. Built by JS — see §7 for the required DOM.

```

.q {
  background: var(--surface);
  border: var(--border);
  box-shadow: var(--sh-sm);
  padding: var(--s-5);
  margin-bottom: var(--s-4);
  position: relative;
}
.q_id {
  position: absolute;
  top: calc(-1 * var(--bw));
  left: calc(-1 * var(--bw));
  background: var(--ink);
  color: var(--paper);
  font-family: var(--font-mono);
  font-size: var(--t-xs);
  font-weight: 700;
  padding: var(--s-1) var(--s-2);
  letter-spacing: 0.08em;
}
.q_text {
  font-size: var(--t-lg);
  font-weight: 500;
  margin: var(--s-4) 0 var(--s-5);
  max-width: 60ch;
}
.q_control { display: flex; align-items: center; gap: var(--s-4); flex-wrap: wrap; }
.q_value {
  font-family: var(--font-mono);
  font-weight: 700;
  font-size: var(--t-2xl);
  min-width: 4.5ch;
  text-align: right;
}
.q_options { display: grid; gap: var(--s-3); margin-bottom: var(--s-4); }
.q_option {
  display: flex; align-items: center; justify-content: space-between;
  gap: var(--s-3);
  border: var(--border-thin);
  background: var(--paper);
  padding: var(--s-2) var(--s-3);
}
.q_sum {
  font-family: var(--font-mono);

```

```
font-weight: 700;  
font-size: var(--t-sm);  
padding: var(--s-2) var(--s-3);  
border: var(--border-thin);  
background: var(--paper);  
}  
.q_sum--bad { background: var(--marvel-yellow); } /* total ≠ 100 */  
.q--answered { box-shadow: var(--sh-xs); transform: var(--press); }  
.q--answered .q_id { background: var(--marvel-green); }
```

6.7 CHARACTER CARD (`.CHAR`) — DRAFT BOARD

```

.char {
  background: var(--surface);
  border: var(--border);
  box-shadow: var(--sh-sm);
  padding: var(--s-4);
  display: flex;
  flex-direction: column;
  gap: var(--s-3);
  min-height: 150px;
  position: relative;
  transition: transform var(--dur-fast) var(--ease),
              box-shadow var(--dur-fast) var(--ease);
}
.char:hover { transform: translate(-2px,-2px); box-shadow: var(--sh); }
.char_name {
  font-family: var(--font-display);
  font-size: var(--t-lg);
  text-transform: uppercase;
  line-height: var(--lh-tight);
}
.char_meta { font-family: var(--font-mono); font-size: var(--t-xs); color:
var(--text-quiet); }
.char button { margin-top: auto; }

/* Taken state – dead card, stamped, no hover */
.char--taken {
  background: var(--paper-dark);
  box-shadow: none;
  transform: var(--press);
}
.char--taken:hover { transform: var(--press); box-shadow: none; }
.char--taken .char_name { color: var(--text-quiet); }
.char_stamp { position: absolute; top: var(--s-4); right: var(--s-3); }

```


6.8 LEADERBOARD TABLE (`.BOARD`)

```
.board { width: 100%; border-collapse: collapse; font-family: var(--font-mono);
}
.board th {
  background: var(--ink);
  color: var(--paper);
  font-family: var(--font-display);
  font-size: var(--t-sm);
  text-transform: uppercase;
  letter-spacing: 0.08em;
  text-align: left;
  padding: var(--s-3) var(--s-4);
  border: var(--border);
}
.board td {
  padding: var(--s-3) var(--s-4);
  border: var(--border);
  font-size: var(--t-lg);
  font-weight: 700;
  background: var(--surface);
}
.board tr:nth-child(even) td { background: var(--paper); }
.board tr[data-rank="1"] td { background: var(--marvel-yellow); }
.board tr[data-rank="2"] td { background: var(--paper-dark); }
.board tr[data-rank="3"] td { background: #E8C9A0; }
.board td:first-child { width: 5rem; font-family: var(--font-display); font-
size: var(--t-xl); }
```

6.9 STEP TRACKER (`.STEP`) — LANDING PAGE ONLY

```
.step { display: grid; grid-template-columns: 72px 1fr; gap: var(--s-4); margin-bottom: var(--s-2); }
.step_num {
  width: 72px; height: 72px;
  display: flex; align-items: center; justify-content: center;
  font-family: var(--font-display);
  font-size: var(--t-2xl);
  background: var(--marvel-red);
  color: var(--paper-bright);
  border: var(--border);
  box-shadow: var(--sh-sm);
}
.step_num--zero { background: var(--ink); }
.step_rail { grid-column: 1; justify-self: center; width: var(--bw);
background: var(--ink); height: var(--s-5); }
.step_time {
  font-family: var(--font-mono);
  font-size: var(--t-xs);
  font-weight: 700;
  text-transform: uppercase;
  letter-spacing: 0.08em;
  color: var(--marvel-red);
  display: block;
  margin-bottom: var(--s-1);
}
```

6.10 TOAST (`.TOAST`) — REPLACES EVERY `ALERT()`

The current code calls `alert()` for errors and confirmations. `alert()` is a browser-chrome modal that cannot be styled and blocks the page — unacceptable in a designed UI. Add a toast utility to `theme.css` and a `toast()` helper in a new `shared/toast.js`.

```

.toast-stack {
  position: fixed;
  bottom: var(--s-5); right: var(--s-5);
  z-index: var(--z-toast);
  display: flex; flex-direction: column; gap: var(--s-3);
  max-width: 380px;
}
.toast {
  background: var(--surface);
  border: var(--border);
  box-shadow: var(--sh);
  padding: var(--s-4);
  font-weight: 500;
  animation: toast-in var(--dur-slow) var(--ease);
}
.toast--ok { border-left: var(--s-2) solid var(--marvel-green); }
.toast--error { border-left: var(--s-2) solid var(--marvel-red); }
.toast--info { border-left: var(--s-2) solid var(--marvel-blue); }
@keyframes toast-in { from { transform: translateX(20px); opacity: 0; } to {
transform: none; opacity: 1; } }

```

```

// portal/public/shared/toast.js
function toast(message, kind = 'info', ms = 4500) {
  let stack = document.querySelector('.toast-stack');
  if (!stack) {
    stack = document.createElement('div');
    stack.className = 'toast-stack';
    stack.setAttribute('role', 'status');
    stack.setAttribute('aria-live', 'polite');
    document.body.appendChild(stack);
  }
  const el = document.createElement('div');
  el.className = `toast toast--${kind}`;
  el.textContent = message;
  stack.appendChild(el);
  setTimeout(() => el.remove(), ms);
}

```

Then replace every `alert(...)` call across `predict.js`, `draft.js`, `report.js`, `judge.js` with `toast(..., 'error')` or `toast(..., 'ok')`. **Keep the message strings exactly as they are** — they were written to be actionable.

6.11 EMPTY / LOADING STATES

Never leave a container blank while data loads. Every async container gets a placeholder.

```
.placeholder {
  border: var(--bw) dashed var(--ink);
  background: var(--paper);
  padding: var(--s-6);
  text-align: center;
  font-family: var(--font-mono);
  font-size: var(--t-sm);
  text-transform: uppercase;
  letter-spacing: 0.1em;
  color: var(--text-quiet);
}
```

Copy, by context:

- Questions loading → RETRIEVING ARCHIVE...
- Draft board loading → SCANNING ROSTER...
- Leaderboard empty → NO STANDINGS YET – ARCHIVE QUIET
- Judge queue done → QUEUE COMPLETE – NO REPORTS PENDING
- Load failure → ARCHIVE UNREACHABLE – CHECK CONNECTION AND RELOAD

7. THE DOM CONTRACT — DO NOT BREAK THIS

Every JS file queries elements by exact ID. If you rename or remove any of these, that page silently stops working, and you will not find out until the event. Restyle freely; never re-key.

You may add wrapper elements, classes, and new markup around these — the requirement is only that an element with the listed ID exists, and (where noted) that a JS-created element keeps its class hooks.

7.1 `EXPLORE.HTML` + `PREDICT.HTML` → `PREDICT.JS`

ID / selector	Role	Notes
<code>#team-id-prompt</code>	Team-ID gate container	JS sets <code>.style.display = 'none'</code>
<code>#team-id-input</code>	Text input	<code>.value</code> read on save
<code>#team-id-save</code>	Button	Click listener attached
<code>#questions</code>	Question list container	JS sets <code>.style.display = 'block'</code> , appends children
<code>window.DOOMSDAY_ROUND</code>	Round filter	<code>explore.html</code> MUST set this to <code>'explore'</code> before loading <code>predict.js</code>
<code>#range-{docId}</code>	Per-question slider	Created by JS
<code>#val-{docId}</code>	Live percentage readout	Created by JS
<code>#submit-{docId}</code>	Per-question submit	Created by JS
<code>#sum-{docId}</code>	Multichoice total readout	Created by JS
<code>.opt-input</code>	Multichoice number inputs	Class is queried — must survive
<code>data-option</code> attr	Option name on each input	Read on submit — must survive

Because JS builds the question markup, restyling the question card requires editing `predict.js`'s template strings. Update the HTML *inside* `predict.js` to the `.q` structure in §6.6, keeping every ID and `.opt-input` / `data-option` hook.

7.2 `DRAFT.HTML` → `DRAFT.JS`

<u>ID / selector</u>	<u>Role</u>
<code>#board</code>	Grid container; JS clears with <code>innerHTML = ''</code> and appends <code>.panel</code> cards
card class <code>panel</code>	Set by JS — change to <code>char</code> in <code>draft.js</code> when restyling
<code>data/draft_pool.json</code>	Fetches at runtime — path must stay relative to the hosting root

7.3 `REPORT.HTML` → `REPORT.JS`

<u>ID</u>	<u>Role</u>
<code>#report-form</code>	Container the three prediction blocks append into
<code>#submit-report</code>	Submit button; listener attached inside <code>loadTopThree()</code>
<code>#chart-{0..2}</code>	Textareas, read on submit
<code>#trap-{0..2}</code>	Textareas, read on submit

Known fragility: `#submit-report` only becomes functional after `loadTopThree()` resolves. If a team has zero submissions, `top3` is empty and the button still binds but submits an empty `entries` array. Leave the behavior as-is; add a `.placeholder` reading `NO PREDICTIONS FOUND — COMPLETE ROUND 1 OR 2 FIRST` when `top3.length === 0`.

7.4 `ADMIN.HTML` → `ADMIN.JS`

<u>ID</u>	<u>Role</u>
<code>#board</code>	<code><table></code> ; JS writes full <code>innerHTML</code> including <code><tr><th></code> header row
<code>#reveal-status</code>	Reveal-state text line

To use the `.board` styling in §6.8 and rank-based row colors, edit `admin.js` to emit `<tr data-rank="{r.rank}">` and give the table `class="board"`.

7.5 `JUDGE.HTML` → `JUDGE.JS`

<u>ID / selector</u>	<u>Role</u>
<code>#judge-id-prompt</code>	<u>Judge-ID gate</u>
<code>#judge-id-input</code>	<u>Text input</u>
<code>#judge-id-save</code>	<u>Button</u>
<code>#report-view</code>	<u>Queue container; JS writes full <code>innerHTML</code></u>
<code>#submit-judge</code>	<u>Created per report by JS</u>
<code>input[type=checkbox]</code> + <code>data-cat</code> + <code>data-pts</code>	<u>Scoring depends on these attributes. Never remove.</u>

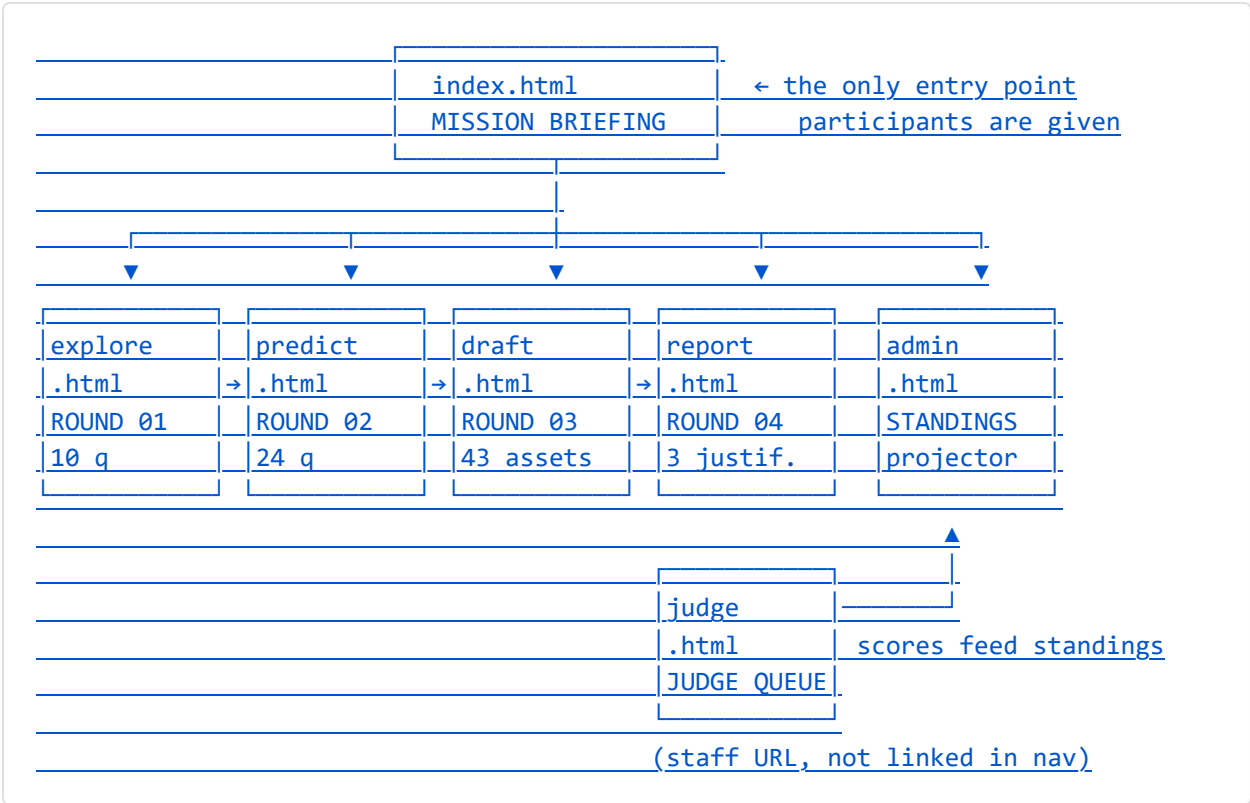
7.6 GLOBAL

<u>Symbol</u>	<u>Source</u>	<u>Note</u>
<code>renderBanner()</code>	<code>shared/banner.js</code>	<u>Prepends <code>.banner-earth4471</code> to <code><body></code></u>
<code>renderNav(activeId)</code>	<code>shared/nav.js</code>	<u>Inserts <code>nav.top-nav</code> as the 2nd body child</u>
<code>firebaseConfig</code>	<code>shared/firebaseConfig.js</code>	<u>Must load before any page script</u>
<code>localStorage['doomsday_team_id']</code>	<u>All participant pages</u>	<u>The only persistence</u>

Nav page IDs (must match `renderNav()` arguments): `landing`, `explore`, `predict`, `draft`, `report`, `leaderboard`.

8. SITE MAP

Seven pages. Five are participant-facing, two are staff-facing.



#	Page	Audience	In nav?	Purpose
1	<code>index.html</code>	Participants	✓ Home	Full briefing + round tracker. The only page that explains the game.
2	<code>explore.html</code>	Participants	✓ Explore	Round 1 — 10 lookup questions
3	<code>predict.html</code>	Participants	✓ Predict	Round 2 — 24 forecast questions
4	<code>draft.html</code>	Participants	✓ Draft	Round 3 — exclusive character claim
5	<code>report.html</code>	Participants	✓ Report	Round 4 — written justification
6	<code>admin.html</code>	Everyone (projector)	✓ Leaderboard	Live rank + total
7	<code>judge.html</code>	Judges only	✗ hidden	Blind scoring queue

`judge.html` is deliberately absent from `nav.js`. It is reachable only by typing the URL. Keep it that way.

9. PAGE SPECIFICATIONS

9.1 `INDEX.HTML` — MISSION BRIEFING

Purpose. There is no live host at this event. This page is the entire briefing. A team must be able to land here cold and understand the game, the rounds, the scoring, and the time budget without asking anyone.

Max width: `--w-content` (960px).

Wireframe:

EARTH-4471 ARCHIVE – This is not Earth-616 ... (magenta)						banner
HOME	EXPLORE	PREDICT	DRAFT	REPORT	LEADERBOARD	nav (sticky)

EARTH-4471 // BRIEFING			win--lift
DOOMSDAY ← .misprint, magenta ghost			
ALGORITHM ← 88px Archivo Black, red			
MCU PREDICTION CHALLENGE • V-TAPP			← .caption
You have four completed phases of a film universe that does not exist. Phase 5 XXXXXXXXXX has not happened. Predict it. ↑ .redact			
09:00	~2.5-3	34	stat trio
-18:00	HOURS	QUESTIONS	
WINDOW	RUNTIME	TOTAL	
[START ROUND 1 →] [DOWNLOAD DATA]			

EARTH-4471 // ARRIVING LATE? (yellow)						win--warn
Start anyway. Rounds 1, 2 and 4 score on submit – partial progress keeps every point it earns.						

0	5 MINUTES
	SET UP
	Pick a team ID • download six files
	films.csv characters.csv appearances.csv ...
1	30-45 MINUTES • 10 QUESTIONS

ROUND 01 – EXPLORE

Answers are already in your files.

[START ROUND 1 →]

2

60-90 MINUTES • 24 QUESTIONS

ROUND 02 – PREDICT

Phase 5. No file has these answers.

[START ROUND 2 →]

3

5 MINUTES • POOL SHRINKS ALL DAY

ROUND 03 – DRAFT

[START ROUND 3 →]

4

20-30 MINUTES

ROUND 04 – REPORT

[START ROUND 4 →]

EARTH-4471 // SCORING

0.5 × ANALYSIS + 0.2 × DRAFT + 0.3 × REPORT

YOU SAID	TRUTH	POINTS	← scoring table
100%	YES	100	
90%	YES	99	
50%	EITHER	75	
90%	NO	19	
100%	NO	0	

Honest beats loud. ← .fx

mono, large

EARTH-4471 // THE ARCHIVE IS NOT CLEAN (glitch)

Three traps are planted in this data. Naive models will find strong, confident, wrong answers. Catching one earns points in Round 4.

▸ SIMPSON'S PARADOX ▸ SURVIVORSHIP ▸ LEAKY COLUMN

win--glitch

(names only – no explanation. Let them find it.)

Content requirements:

- The stat trio numbers **MUST** render in `--font-mono`, `--t-3x1`, **bold**.
- The six CSV links live inside the Step 0 window, styled as a `.files` row: `mono`, `--marvel-blue`, underlined, separated by `.`.
- Every “START ROUND N” is a `.btn--lg`.
- **The traps window names the three traps but never explains them.** Naming them is a fair warning; explaining them gives away Round 4 points.
- No countdown timer. There is no shared clock in a drop-in event; a ticking timer would be fiction.

States: none. This page is fully static — no Firestore reads, no JS beyond `renderBanner()` / `renderNav('landing')` and the optional headline animation.

Optional motion (§12.3): the word after “PREDICT THE” cycles through `UNWRITTEN` / `UNRELEASED` / `UNKNOWABLE`. 15 lines of vanilla JS, no library. Skip it if time is short.

9.2 `EXPLORE.HTML` — ROUND 01

Purpose. Ten questions whose answers are already in the downloaded CSVs. This round exists to force an actual read of the data.

Max width: `--w-content`.

Wireframe:

banner / nav

EARTH-4471 // ROUND 01 – EXPLORE

ROUND 01 ← .caption
EXPLORE ← h1

Every answer is already in the files you downloaded.
Nothing here is a prediction.

10

30-45

GRADED

← stat chips

QUESTNS

MINUTES

ON

SUBMIT

FILES YOU NEED:
films.csv · characters.csv · appearances.csv ·
post credits.csv ← mono, blue

TEAM GATE (#team-id-prompt)

EARTH-4471 // IDENTIFY (yellow)

TEAM ID

[] [CONTINUE]

Use the exact same ID on every page.

— after gate passes, #questions becomes visible —

[e1]

Did Earth-4471 Chronicle 4 earn a higher worldwide gross than Earth-4471 Chronicle 19?

NO YES 62 %

[SUBMIT]

[e5]

Which of these has the most total screentime across Phases 1-4?

	Nick Fury	[25] %
	Riri Williams	[25] %
	Loki	[25] %
	Okoye	[25] %
	TOTAL: 100%	[SUBMIT]

Check the projector for live standings. ← quiet note

Question card details:

- `.q_id` badge shows the raw question ID (`e1`, `e5`) — mono, black tab in the top-left corner, overlapping the border.
- Slider track is labeled `NO` at 0 and `YES` at 100, both in mono `--t-xs`.
- `.q_value` shows the live percentage at `--t-2x1` mono, updating on `input`.
- Multichoice options render as `.q_option` rows with the option name left, number input right.
- `.q_sum` turns `--marvel-yellow` when the total $\neq$ 100 (advisory only — do not block submission; the scorer normalizes).

States:

<u>State</u>	<u>Visual</u>
<u>Loading</u>	<code>#questions</code> shows <code>.placeholder</code> → <code>RETRIEVING ARCHIVE...</code>
<u>Gate not passed</u>	<code>#questions</code> hidden, <code>#team-id-prompt</code> visible
<u>Gate passed</u>	<code>#team-id-prompt</code> hidden, <code>#questions</code> visible
<u>Question answered</u>	Button → <code>SUBMITTED ✓</code> , <code>.btn--done</code> green; card gets <code>.q--answered</code> (presses into its shadow, ID badge turns green)
<u>Submit failed</u>	<code>toast(msg, 'error')</code> — card stays interactive
<u>Load failed</u>	<code>.placeholder</code> → <code>ARCHIVE UNREACHABLE – CHECK CONNECTION AND RELOAD</code>

Progress affordance (add this). A sticky bar under the nav: `ROUND 01 ▸ 4 / 10 ANSWERED`, mono, on `--marvel-yellow`, full-width, `border-bottom: var(--border)`. Count is derived client-side from how many buttons carry `.btn--done`. Purely local — **no extra Firestore reads.**

9.3 `PREDICT.HTML` — ROUND 02

Structurally identical to Explore. Every difference is copy and emphasis.

	<u>Explore</u>	<u>Predict</u>
<u>Title bar</u>	EARTH-4471 // ROUND 01 – EXPLORE	EARTH-4471 // ROUND 02 – PREDICT [SEALED]
<u>Window variant</u>	default	.win--glitch on the intro window
<u>Stat chips</u>	10 · 30–45 min · graded on submit	24 · 60–90 min · Brier scored
<u>Framing line</u>	"Every answer is already in your files."	"No file contains these answers."
<u>Extra callout</u>	"Files you need: ..."	Brier scoring mini-table + trap warning
<u>Question ID badges</u>	e1 – e10	q1 – q24
<u>Phase 5 mentions</u>	—	Rendered with .stamp--sealed beside the heading

Add to the Predict intro window (Explore does not get this):

HOW CONFIDENCE IS SCORED
YOU SAY 100% → RIGHT: 100 pts WRONG: 0 pts
YOU SAY 90% → RIGHT: 99 pts WRONG: 19 pts
YOU SAY 50% → EITHER WAY: 75 pts
Skipping = 50%. An honest shrug costs nothing extra.

Because 24 questions is a long scroll, **group them visually**: insert a section divider window before each block of question types.

== SURVIVAL · 10 QUESTIONS ==
== TEAM-UPS · 6 QUESTIONS ==
== CLOSEST PARTNER · 4 QUESTIONS ==
== SCREENTIME · 4 QUESTIONS ==

Dividers are derived client-side by inspecting question text — **do not add fields to Firestore for this**. Match on the question text: contains `survive Phase 5` → Survival; `team-up` → Team-ups; `share the most scenes` → Closest partner; `more total screentime` → Screentime. If a match fails, render the question with no divider rather than crashing.

9.4 `DRAFT.HTML` — ROUND 03

Purpose. Claim one character, exclusively, before someone else does. The only round with genuine time pressure.

Max width: `--w-wide` (1240px).

Wireframe:

banner / nav

EARTH-4471 // DRAFT BOARD – 43 ASSETS		
ROUND 03		
DRAFT		
Claim ONE character you believe survives Phase 5.		
First team to claim takes them off the board		
for everyone else. Permanently.		
31	12	1
AVAILABLE	CLAIMED	PICK EACH
↑ live counts from the snapshot		

MANTIS	LOKI	SHURI	THANOS	VISION
			/TAKEN\	
[DRAFT]	[DRAFT]	[DRAFT]	dimmed	[DRAFT]
...43 cards total, 5 across on desktop				

Grid: `repeat(auto-fill, minmax(200px, 1fr)) , gap --s-4` . Do not hard-code 4 columns — 43 cards at fixed columns leaves an ugly orphan row on most widths.

Card states:

<u>State</u>	<u>Visual</u>
<u>Available</u>	<code>.char</code> , white surface, <code>[DRAFT]</code> red button, lifts on hover
<u>Taken by another team</u>	<code>.char--taken</code> : grey fill, no shadow, pressed-in, <code>TAKEN</code> stamp rotated 4°, name in <code>--text-quiet</code> , no button
<u>Taken by <i>this</i> team</u>	<code>.char--taken</code> + <code>.stamp--done</code> reading <code>YOURS</code> , green
<u>This team already picked</u>	All remaining cards lose their button; a sticky yellow bar reads <code>YOU HAVE DRAFTED – ONE PICK PER TEAM</code>
<u>Pool empty</u>	<code>.placeholder</code> → <code>ROSTER EXHAUSTED – ALL 43 ASSETS CLAIMED</code>

Live updates. `draft_picks` has an `onSnapshot` listener. A card flipping to taken while the participant watches is the single most dramatic moment in the UI — **give it a transition:** 200ms, card fills grey, stamp scales in from 1.4× to 1×. Do not animate all 43 on first paint (see §12.2).

Critical: when restyling, `draft.js` currently sets `card.className = 'panel'`. Change it to `'char'` and rebuild the inner markup to the `.char` structure. Keep the `#board` container ID and the batch-write logic exactly as they are.

9.5 `REPORT.HTML` — ROUND 04

Purpose. Written justification of the team's three boldest predictions. Judged by hand.

Max width: `--w-narrow` (720px) — this is a writing page; long lines hurt.

Wireframe:

banner / nav

 EARTH-4471 // ROUND 04 – JUSTIFY
ROUND 04
DEFEND YOUR CALLS
We pulled your three boldest predictions. For each, say what in the data backs it – and name any trap you caught.
WHAT EARNS POINTS (yellow)
▸ Cite a specific chart or number, not a vibe
▸ Combine more than one CSV
▸ Name a trap and say how you handled it
▸ Half a page. A stranger should follow it in under a minute.

PREDICTION 1
QUESTION q7 ← mono badge
YOU SAID: 92% ← mono, red, big
CHART / NUMBER THAT BACKS IT
TRAP NOTICED, AND HOW YOU HANDLED IT

PREDICTION 2 (same)

PREDICTION 3 (same)

[SUBMIT REPORT] ← .btn--lg, block

Details:

- Each prediction block is a `.win` with title bar `PREDICTION N`.
- `YOU SAID: 92%` in mono, `--t-2x1`, `--marvel-red`. This is the team's own boldness reflected back — make it loud.
- Textareas: `min-height: 96px`, mono font (they're citing data).
- **Empty state:** if `top3.length === 0`, replace the form with `.placeholder` → `NO PREDICTIONS FOUND — COMPLETE ROUND 1 OR 2 FIRST`, and disable `#submit-report`.
- **Submitted state:** after a successful write, replace the whole form with a success window: green title bar, `REPORT LOGGED`, plus `A judge will score it. You don't have to wait.` Do not leave the form editable — it writes to `reports/{teamId}` and a resubmit silently overwrites.

9.6 `ADMIN.HTML` — LIVE STANDINGS (PROJECTOR)

Purpose. Runs on a projector for hours. Read from across a room. This is the one page that should be **dark** — a lit screen in a bright hall, and it is the visual payoff of the whole event.

Max width: `--w-wide`. No nav interaction needed, but keep the nav for organizer convenience.

Inverted palette (scoped to this page only):

```
body.projector {
  --bg: var(--ink);
  --surface: #16130F;
  --text: var(--paper);
  --text-quiet: #8A8279;
  background: var(--ink);
}
body.projector .win { border-color: var(--paper); box-shadow: 10px 10px 0 var(--marvel-red); }
body.projector .win bar { background: var(--marvel-red); color: var(--paper-bright); border-bottom-color: var(--paper); }
body.projector .board th { background: var(--paper); color: var(--ink); border-color: var(--paper); }
body.projector .board td { background: var(--surface); color: var(--paper); border-color: var(--paper); }
body.projector .board tr:nth-child(even) td { background: #1E1A15; }
body.projector .board tr[data-rank="1"] td { background: var(--marvel-yellow); color: var(--ink); }
```

Wireframe:

EARTH-4471 ARCHIVE – ...

LIVE STANDINGS

← .misprint, huge, paper on ink

PHASE 5 NOT YET REVEALED

← #reveal-status
as .stamp--sealed

EARTH-4471 // STANDINGS [LIVE]

RANK	TEAM	TOTAL	
1	QUANTUM CORE	84.2	← yellow row
2	NULL POINTER	81.7	← grey row
3	VARIANCE	79.0	← bronze row
4	SIGMA SQUAD	76.4	
5	...		

Requirements:

- Base font size on this page: `--t-x1` **minimum**. Rank column at `--t-3x1`. This is read from 10 metres away.
- Rank 1 row: `--marvel-yellow` background, ink text, and a `.fx` `#1` marker in the margin.
- `#reveal-status` renders as a `.stamp`: `.stamp--sealed` (magenta) while unrevealed, `.stamp--done` (green, reading `PHASE 5 REVEALED – DRAFT FINAL`) after.
- **Empty state:** `.placeholder` → `NO STANDINGS YET – ARCHIVE QUIET`. This is what the room sees at 9:00 AM; make it look intentional, not broken.
- **Rank-change motion:** when a row's rank improves, flash its background to `--marvel-yellow` for 600ms then settle. Track previous ranks in a local object. Do not animate on the first snapshot.
- **Never show per-round breakdowns.** The `leaderboard_public` collection only carries `rank` and `total` by deliberate design — a visible raw Predict score is an answer-extraction oracle. If you find yourself adding columns, stop.

Auto-refresh safety: this page sits open for 9 hours. Ensure no unbounded array grows across snapshots (rebuild the table body each time, which the current code already does).

9.7 `JUDGE.HTML` — JUDGE QUEUE

Purpose. A judge works through submitted reports blind and ticks a checklist. Target: 60–90 seconds per report. **Speed and scannability beat beauty here.**

Max width: `--w-narrow`.

Wireframe:

banner (no nav – this page is not in the participant flow)

#judge-id-prompt	
 EARTH-4471 // JUDGE ACCESS	(yellow)
JUDGE ID	
[]	[START]
Reports are shown blind. Team names are hidden.	

— after start, #report-view —

 EARTH-4471 // REPORT 3 OF 11
q7 • "Screentime dropped every phase, but within Phase 3 alone the correlation flips positive..."
TRAP: "Simpson's paradox – segmented by phase"
(one bordered block per prediction, mono, escaped)

1 • DATA SOPHISTICATION	— / 25
☐ Combined more than one data table	+10
☐ Used a real statistic – rate/trend/corr	+10
☐ Tool/method matches the claim	+5
RUNNING: 20 / 25	

2 • LOGIC & EVIDENCE	/25	(4 checks)
3 • VISUALIZATION	/25	(3 checks)
4 • COMMUNICATION	/25	(3 checks)

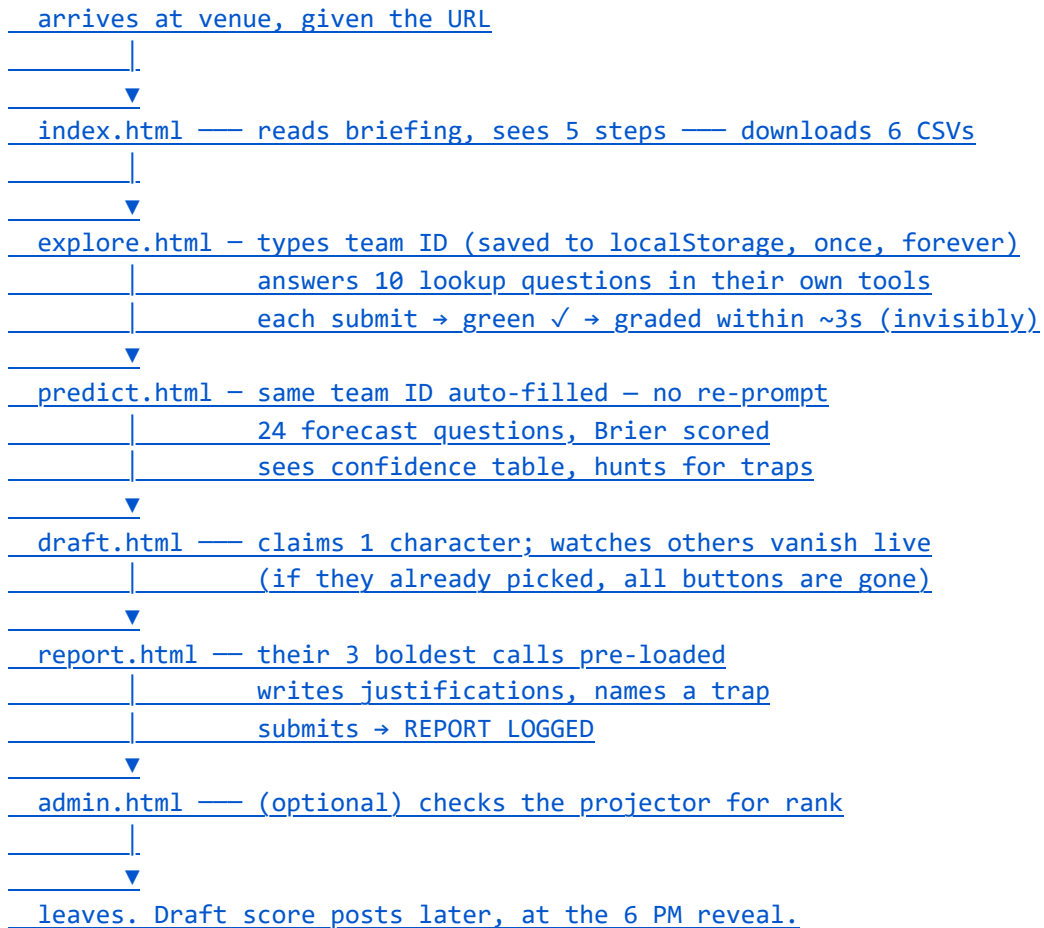
TOTAL	68 / 100	sticky
[SUBMIT AND NEXT →]		bottom

Requirements:

- **Add a live running total.** The current code computes the total only on submit. A judge should see the number they're building. Implement with an `input` listener on `#report-view` that sums `:checked` boxes' `data-pts` — **read the same attributes the submit handler uses**, so the two can never disagree.
 - The total bar is `position: sticky; bottom: 0; --marvel-yellow` background, `border-top: var(--border-thick)`, total in mono at `--t-2x1`.
 - Checkboxes are 24px (§6.5). Each check row is a `<label>` with `display: flex; gap: var(--s-3); padding: var(--s-2)`, and the whole row is clickable. `cursor: pointer` on the label.
 - Point values right-aligned in mono, `--marvel-blue`.
 - Team-authored text (`chartJustification`, `trapNote`) renders in a bordered mono block. **It is already HTML-escaped in `judge.js` — do not remove `escapeHtml()`**. That escaping is a security control: a judge's session can create `judge_scores` documents.
 - **Queue complete:** `.placeholder` → `QUEUE COMPLETE — NO REPORTS PENDING`, plus a `.fx` reading `NICE`.
 - **Do not display the team ID anywhere.** The queue is blind by design; only the position (`REPORT 3 OF 11`) is shown.
-

10. USER FLOWS

10.1 PARTICIPANT (THE 2.5–3 HOUR PATH)



Persistence rule: the team ID is typed **once**. `localStorage['doomsday_team_id']` is read by every subsequent page. If a team switches laptops they must retype it — the landing page says so.

Failure paths:

- Lands on `report.html` first → `prompt()` fallback for team ID, then `NO PREDICTIONS FOUND` placeholder.
- Lands on `draft.html` first → `prompt()` fallback, draft works fine standalone.

- Refreshes mid-round → answered questions do **not** show as answered (submissions aren't re-read on load). This is known and accepted; a resubmit simply overwrites the same document ID. **Do not “fix” this by adding a per-load submissions query** — that multiplies Firestore reads across every team for cosmetic benefit.

10.2 ORGANIZER

```

before 09:00 — node grading-server.js (terminal stays open all day)
| confirms "Loaded 34 questions ... Answer key: present"
| opens admin.html on the projector
▼
09:00-18:00 — nothing. Watches standings. Assigns team IDs at check-in.
|
▼
~18:00 — second terminal: node reveal.js
| #reveal-status flips to PHASE 5 REVEALED
| every team's draft score posts at once
▼
announces winners from the projector

```

10.3 JUDGE

```

given judge.html URL (not linked anywhere)
|
▼
types Judge ID — queue loads all submitted reports, blind
|
▼
per report: read 3 justifications → tick boxes → watch running total
| → SUBMIT AND NEXT (60-90 seconds)
▼
QUEUE COMPLETE — scores average server-side into each team's total

```

Multiple judges can work the same queue simultaneously under different Judge IDs. The document key is `{judgeId}_{reportId}`, so two judges never collide, and the grading server averages them.

11. RESPONSIVE

Three breakpoints. No more.

```
/* Mobile-first base: single column, everything full width */

@media (min-width: 720px) {
  /* stat chips go horizontal; draft board 3 across */
}
@media (min-width: 1024px) {
  /* draft board auto-fill; landing steps get the number rail */
}
@media (min-width: 1440px) {
  /* projector sizing on admin.html only */
}
```

Per-page rules:

<u>Page</u>	<u>< 720px</u>	<u>720–1024px</u>	<u>> 1024px</u>
<u>index</u>	<u>Step numbers shrink to 48px, sit above the card</u>	<u>Full step layout</u>	<u>Full</u>
<u>explore / predict</u>	<u>Slider full width, value below</u>	<u>Side-by-side</u>	<u>Side-by-side</u>
<u>draft</u>	<u>2 cards across</u>	<u>3 across</u>	<u>auto-fill, 5–6 across</u>
<u>report</u>	<u>Single column (already)</u>	<u>Single</u>	<u>Single, 720px max</u>
<u>admin</u>	<u>Rank + total only, hide long names with ellipsis</u>	<u>Full table</u>	<u>Projector scale</u>
<u>judge</u>	<u>Checks stack, sticky total stays</u>	<u>Full</u>	<u>Full</u>

Shadow scaling: below 720px, reduce `--sh` to `4px 4px 0` and `--sh-1g` to `6px 6px 0`. A 10px offset shadow on a 360px screen eats the layout.

```
@media (max-width: 719px) {
  :root { --sh: 4px 4px 0 var(--ink); --sh-lg: 6px 6px 0 var(--ink); --sh-xl:
8px 8px 0 var(--ink); }
}
```

Nav on mobile: horizontally scrollable (`overflow-x: auto`, already set). Do not build a hamburger menu — six items, and a burger hides the round structure that is the whole point.

12. MOTION

Motion is **mechanical, fast, and physical**. Nothing floats, nothing fades slowly, nothing eases in a bouncy way. The whole vocabulary is: things move into their shadows, things stamp into place, things flash once.

12.1 INTERACTION MOTION (REQUIRED)

Element	Trigger	Motion
Button	<code>:active</code>	<code>translate(3px, 3px)</code> + shadow 4px→2px, 80ms
Character card	<code>:hover</code>	<code>translate(-2px, -2px)</code> + shadow 4px→6px, 80ms
Nav link	<code>:hover</code>	background → yellow, 80ms, no movement
Question card	on submit	gains <code>.q--answered</code> ; presses in, ID badge → green, 140ms
Toast	on create	slide 20px from right + fade, 260ms
Slider thumb	drag	none (native)

12.2 SNAPSHOT MOTION (CAREFUL)

Firestore `onSnapshot` fires once on attach with **every existing document as an “added” change**. Animating that first paint means 43 draft cards or 50 leaderboard rows all animating at once on load — a mess, and janky on a projector left running for

[hours.](#)

Rule: track a `firstPaint` boolean per listener. Skip all entrance animation while it is true; set it false after the first snapshot completes.

```
let firstPaint = true;
db.collection('draft_picks').onSnapshot(snap => {
  renderBoard(buildTakenMap(snap), { animate: !firstPaint });
  firstPaint = false;
});
```

<u>Event</u>	<u>Motion (after first paint only)</u>
Character becomes taken	Card greys , <code>TAKEN</code> stamp scales 1.4→1, <code>200ms</code>
Leaderboard rank improves	Row background flashes yellow , <code>600ms</code> , then settles
Reveal flips true	<code>#reveal-status</code> stamp swaps sealed→done with a single 300ms scale pop

12.3 OPTIONAL: LANDING HEADLINE CYCLE

```
// index.html, inline. Cycles the word after "PREDICT THE".
(function () {
  const el = document.getElementById('cycle-word');
  if (!el) return;
  const words = ['UNWRITTEN', 'UNRELEASED', 'UNKNOWABLE'];
  let i = 0;
  setInterval(() => {
    i = (i + 1) % words.length;
    el.style.transform = 'translateY(-100%)';
    el.style.opacity = '0';
    setTimeout(() => {
      el.textContent = words[i];
      el.style.transition = 'none';
      el.style.transform = 'translateY(100%)';
      requestAnimationFrame(() => {
        el.style.transition = 'transform 260ms cubic-bezier(0.2,0,0,1), opacity 260ms';
        el.style.transform = 'translateY(0)';
        el.style.opacity = '1';
      });
    }, 260);
  }, 2600);
})();
```

This is the one idea worth keeping from the React reference. It needs no framer-motion.

12.4 REDUCED MOTION

```
@media (prefers-reduced-motion: reduce) {
  *, *::before, *::after {
    animation-duration: 0.01ms !important;
    animation-iteration-count: 1 !important;
    transition-duration: 0.01ms !important;
  }
}
```

The press effect on `:active` may stay — it is a state change, not an animation, and it is the primary affordance.

13. ACCESSIBILITY & PERFORMANCE

13.1 ACCESSIBILITY FLOOR

- **Contrast:** `--ink` on `--paper` = 15.8:1. `--paper-bright` on `--marvel-red` = 4.9:1 — passes AA for the ≥14px bold button text we use, and nothing smaller. Never put `--marvel-yellow` text on paper.
- **Focus:** every interactive element gets `outline: 3px solid var(--marvel-blue); outline-offset: 3px` on `:focus-visible`. Do not remove outlines.
- **Labels:** every input has a real `<label>`. The judge checklist rows are already labels — keep the input *inside* the label so the whole row is clickable.
- **Live regions:** the toast stack is `role="status" aria-live="polite"`. The leaderboard table gets `aria-live="polite"` so rank changes are announced.
- **Redaction bars** keep their text content (invisible via matching color) so screen readers still read the value — this is decorative redaction, not real secrecy. Never use `.redact` for actual secrets; the answer key's secrecy is enforced by Firestore rules, not CSS.
- **Sliders:** add `aria-label` describing the question, and `aria-valuetext="62 percent"` updated alongside the visible value.

13.2 PERFORMANCE BUDGET

This runs on student laptops over campus wifi for three hours.

<u>Item</u>	<u>Budget</u>	<u>Note</u>
<u>Total CSS</u>	<u>< 30 KB</u>	<u>One file, no framework</u>
<u>Fonts</u>	<u>4 families,</u> <u>display=swap</u>	<u>~120 KB — the largest cost, and worth it</u>
<u>JS added by redesign</u>	<u>< 5 KB</u>	<u>toast.js only</u>
<u>Images</u>	<u>zero</u>	<u>Every texture is CSS</u>
<u>Video</u>	<u>zero</u>	<u>The reference's 2.3 MB video is deleted, not adapted</u>
<u>Extra Firestore reads</u>	<u>zero</u>	<u>Redesign must not add a single query</u>

The zero-extra-reads rule is hard. The Spark free tier allows 50,000 reads/day. Current per-team load is ~34 (questions) + ~43 (draft pool via static JSON, no reads) + a handful. A “show which questions I already answered on reload” feature would add ~34 reads per page load per team — tempting, and not worth the quota risk mid-event.

14. IMPLEMENTATION ORDER

Do these in order. Each step leaves the site deployable and working.

Step 1 — Foundation. Rewrite `shared/theme.css` with §3 tokens, §5 typography, §4 textures. Add the font `<link>` to all 7 pages. Deploy. Site will look half-finished but must not be broken. *Verify: every page still loads, no console errors.*

Step 2 — Global chrome. Style `.banner-earth4471` (§6.2) and `nav.top-nav` (§6.3). Do not edit `banner.js` or `nav.js` beyond adding `explore` if missing. *Verify: nav active state correct on all 6 linked pages.*

Step 3 — Component library. Add every component in §6 to `theme.css`: `.win`, buttons, inputs, `.q`, `.char`, `.board`, `.step`, `.toast`, `.placeholder`. Nothing uses them yet. *Verify: CSS parses, no page regressions.*

Step 4 — `index.html` . Full rebuild to §9.1. Pure markup work, no JS risk. This is the highest-visibility page — get it right. *Verify: screenshot at 360px, 768px, 1440px.*

Step 5 — `toast.js` . Add `shared/toast.js` , include it on `predict.html` , `explore.html` , `draft.html` , `report.html` , `judge.html` . Replace every `alert()` with `toast()` , preserving message strings. *Verify: trigger a failed submit, confirm toast appears and auto-dismisses.*

Step 6 — Question rounds. Update the template strings inside `predict.js` to the `.q` markup (§6.6), **preserving every ID and the `.opt-input` / `data-option` hooks (§7.1)**. Rebuild `explore.html` and `predict.html` intro windows. Add the sticky progress bar. *Verify: submit one question on each round on the live site; confirm the leaderboard row appears in Firestore.*

Step 7 — `draft.html` . Rebuild the intro window; change `draft.js` 's `card.className` from `'panel'` to `'char'` and rebuild card markup. Add live available/claimed counts. Add first-paint guard. *Verify: draft a character in one browser, confirm it flips to TAKEN in a second browser within ~1s.*

Step 8 — `report.html` . Rebuild to §9.5, including empty state and submitted state. *Verify: submit a report, confirm `reports/{teamId}` written.*

Step 9 — `admin.html` . Add `class="projector"` to `<body>` , apply the inverted palette, update `admin.js` to emit `class="board"` and `data-rank` . Add rank-change flash with first-paint guard. *Verify: leave open 10 minutes with a live submission; no memory growth, no duplicated rows.*

Step 10 — `judge.html` . Rebuild to §9.7, add the live running total, sticky total bar. **Confirm `escapeHtml()` is still called on both team-authored fields.** *Verify: score a test report end-to-end, confirm `judge_scores` document and `reportRaw` on the leaderboard.*

Step 11 — Full pass. Run every page through the §15 checklist. Delete test data from Firestore. Redeploy.

15. ACCEPTANCE CHECKLIST

Design:

- ☐ No `border-radius` anywhere except `0`
- ☐ Every shadow is `Npx Npx 0` — zero blur, zero spread
- ☐ No `backdrop-filter`, no translucent panels, no gradients on surfaces
- ☐ No color uses `opacity` for emphasis — quiet text uses `--text-quiet`
- ☐ Every content block is a `.win` with a title bar reading `EARTH-4471 // ...`
- ☐ Every scored number renders in `--font-mono`, bold
- ☐ `.misprint` used on exactly two headlines (landing, leaderboard)
- ☐ `.fx` used at most once per page
- ☐ Zero image files, zero video files added

Function (test against the **live deployed site**, not locally):

- ☐ All 7 pages load with zero console errors
- ☐ `explore.html` shows exactly 10 questions; `predict.html` exactly 24
- ☐ A submission on each round writes to `submissions` and produces a `leaderboard` row within ~5s
- ☐ Drafting in browser A flips the card to TAKEN in browser B
- ☐ A second draft attempt by the same team is refused
- ☐ Report page pre-loads the team's three boldest predictions
- ☐ Judge queue scores a report and writes `judge_scores`
- ☐ Leaderboard shows rank + total only — **no per-round columns**
- ☐ `judge.html` is not linked from `nav.js`
- ☐ `escapeHtml()` still called on `chartJustification` and `trapNote`
- ☐ Every `localStorage` read is wrapped or falls back to `prompt()`
- ☐ Zero new Firestore queries added vs. the pre-redesign code

Responsive & a11y:

- ☐ No horizontal body scroll at 360px on any page
- ☐ Tab order reaches every control; focus ring visible on all
- ☐ `prefers-reduced-motion` honored

- ☐ [Leaderboard readable from across a room at projector size](#)
-

16. DO-NOT LIST

1. [Do not migrate to React, Next, Vite, or Tailwind. §1.](#)
2. [Do not rename or remove any ID in §7. The event breaks silently.](#)
3. [Do not add Firestore reads. §13.2.](#)
4. [Do not add per-round score columns to the leaderboard. It is an answer-extraction oracle — a team submits 100% on a throwaway ID, reads the score back, and recovers the answer key one question at a time.](#)
5. [Do not remove `escapeHtml\(\)` from `judge.js`. Team-authored text reaches a privileged session.](#)
6. [Do not link `judge.html` from the nav.](#)
7. [Do not add a background video. The reference has one; it costs 2.3 MB over shared campus wifi for eight seconds of viewing.](#)
8. [Do not explain the three traps on the landing page. Name them; that is the fair warning. Explaining them gives away Round 4 points.](#)
9. [Do not add a countdown timer. Drop-in event, no shared clock — a timer would be fiction.](#)
10. [Do not soften the aesthetic. The instinct after two hours of hard borders is to round a corner or blur a shadow. Resist it. Half-brutalism reads as a mistake; full brutality reads as a decision.](#)